



## AQ1.2: Activity Questions 2 - Not Graded



This assignment will not be graded and is only for practice.

### 2.1 Level 1:

1 point

Suppose  $A = \begin{bmatrix} 2 & 4 & 5 & 1 \\ 11 & -29 & -6 & -3 \\ -3 & 4 & 7 & 7 \end{bmatrix}$ . Which of the following is true about the matrix  $A$

A?

[Hint: The  $(i, j)$ -th entry is the entry which is at the  $i$ -th row and  $j$ -th column.]

☐

It is a  $4 \times 3 \times 4$  matrix.

☒

It is a  $3 \times 4 \times 3$  matrix.

☐

$(2, 3)$ -th entry of the matrix  $AA$  is 44.

☒

$(2, 3)$ -th entry of the matrix  $AA$  is 99.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

It is a  $3 \times 4 \times 3$  matrix.

$(2, 3)$ -th entry of the matrix  $AA$  is 99.

1 point

Which of the following statements is(are) TRUE?

[Hint: Recall, the definitions of scalar matrix, diagonal matrix, and identity matrix.]

☐

Any diagonal matrix is a scalar matrix.

☐

Scalar matrices may not be square matrices.

☒

Scalar matrices must be square matrices.

☐

Any scalar matrix is an identity matrix.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

Scalar matrices must be square matrices.

1 point

Given below is the system of linear equations:

$$7x_1 + 10x_2 + 12x_3 = 368x_1 + 4x_2 - 9x_3 = 114x_1 - x_2 + 3x_3 = 107x_1 + 10x_2 + 12x_3 = 368x_1 + 4x_2 - 9x_3 = 114x_1 - x_2 + 3x_3 = 10$$

If the matrix representation of the system of equations is  $Ax = b$ , where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , then

☐

, then

☒


$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 \\ 8 & 4 & -9 & & \\ 4 & -1 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & & & & \\ 3 & 6 & 1 & 1 & 1 & 0 & & & & & & \end{bmatrix}$$

○

$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 & 3 & 6 \\ 8 & 4 & -9 & 1 & 1 & 1 & \\ 4 & -1 & 3 & 1 & 0 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & 1 & 1 & 2 & 9 & 3 & 6 & 1 & 1 & 1 & 0 & & \\ & & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & & & & & & \\ 3 & 6 & 1 & 1 & 1 & 0 & & & & & & & & & & & \end{bmatrix}$$

○

$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 & 3 & 6 \\ 8 & 4 & -9 & 1 & 1 & 1 & \\ 4 & -1 & 3 & 1 & 0 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & 6 & 1 & 1 & 1 & 0 & & \\ & & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & & & & & & \\ 3 & 6 & 1 & 1 & 1 & 0 & & & & & & & & & & & \end{bmatrix}$$

○

$$A = \begin{bmatrix} 7 & 1 & 0 & 3 & 6 & 1 & 2 \\ 8 & 4 & -9 & 1 & 1 & -9 & \\ 4 & -1 & 1 & 0 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 3 & 6 & 1 & 1 & 0 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & & & & & & \\ 3 & 6 & 1 & 1 & 1 & 0 & & & & & & & & & & & \end{bmatrix}$$

**Yes, the answer is correct.****Score: 1****Accepted Answers:**

$$A = \begin{bmatrix} 7 & 1 & 0 & 1 & 2 \\ 8 & 4 & -9 & & \\ 4 & -1 & 3 & & \end{bmatrix}, b = \begin{bmatrix} 36 \\ 11 \\ 10 \end{bmatrix} A = \begin{bmatrix} & & & & \\ 7 & 8 & 4 & 1 & 0 & 4 & -1 & 1 & 2 & -9 & 3 & & \\ & & & & & & & & & & & \end{bmatrix}, b = \begin{bmatrix} & & & & \\ 3 & 6 & 1 & 1 & 1 & 0 & & & & & & \end{bmatrix}$$

*1 point*

Which of the following statements is(are) TRUE?

☐ Addition of two matrices is possible only if the number of columns in the first matrix is same as the number of rows in the second matrix.

☒ Addition of two matrices is possible only if the orders of both the matrices are the same.

☐

Defining  $ABAB$  is possible if the number of rows in the matrix  $AA$  is same as the number of columns in the matrix  $BB$ .

☒

Defining  $ABAB$  is possible if the number of columns in the matrix  $AA$  is same as the number of rows in the matrix  $BB$ .

**Yes, the answer is correct.****Score: 1****Accepted Answers:**

Addition of two matrices is possible only if the orders of both the matrices are the same.

Defining  $ABAB$  is possible if the number of columns in the matrix  $AA$  is same as the number of rows in the matrix  $BB$ .

*1 point*

Suppose  $P = \begin{bmatrix} 3 & -17 \\ 4 & 0 \\ 2 & -52 \end{bmatrix}$ ,  $Q = \begin{bmatrix} 14 & -9 \end{bmatrix}$ ,  $R = \begin{bmatrix} 0 & -310 \end{bmatrix}$ ,  $D = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$

$D = \begin{bmatrix} \phantom{0} \\ \phantom{0} \\ -245 \end{bmatrix}$

[Hint: If  $AA$  is a matrix of order  $m \times nm \times n$  and  $BB$  is a matrix of order  $n \times pn \times p$ , then the order of  $ABAB$  is  $m \times pm \times p$ .]



The matrix  $PDPD$  is of order  $3 \times 1.3 \times 1$ .



The matrix  $PDPD$  is of order  $1 \times 3.1 \times 3$ .



The matrix  $QDQD$  is of order  $3 \times 3.3 \times 3$ .



The matrix  $QDQD$  is of order  $1 \times 1.1 \times 1$ .



The matrix  $DQDQ$  is of order  $3 \times 3.3 \times 3$ .



The matrix  $DQDQ$  is of order  $1 \times 1.1 \times 1$ .



The product  $QDQD$  is not defined.



The product  $QRQR$  is not defined.



The addition  $P + QP + Q$  is not defined.



The addition  $P + DP + D$  is not defined.

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The matrix  $PDPD$  is of order  $3 \times 1.3 \times 1$ .

The matrix  $QDQD$  is of order  $1 \times 1.1 \times 1$ .

The matrix  $DQDQ$  is of order  $3 \times 3.3 \times 3$ .

The product  $QRQR$  is not defined.

The addition  $P + QP + Q$  is not defined.

The addition  $P + DP + D$  is not defined.

## 2.2 Level 2:

1 point

Choose the set of correct options.

[Hint: Assume  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  and use the given conditions to find matrix  $AA$ , for

first two options.]



If  $AA$  is a square matrix of order 2 and  $A^2 = IA^2 = I$ , then  $A = IA = I$  or  $A = -IA = -I$  where  $II$  is the identity matrix of order 2.



If  $AA$  is a square matrix of order 2 and  $A^2 = 0A^2 = 0$ , then  $A = 0A = 0$ .



If  $AA$  and  $BB$  are square matrices of order 3 and  $A + B = 0A + B = 0$ , then  $B = -AB = -A$ .



If  $AA$  is a scalar matrix of order 3,  $BB$  is a non-zero square matrix of order 3 and  $AB = 0AB = 0$ , then  $A = 0A = 0$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

If  $AA$  and  $BB$  are square matrices of order 3 and  $A + B = 0A + B = 0$ , then  $B = -AB = -A$ .

If  $AA$  is a scalar matrix of order 3,  $BB$  is a non-zero square matrix of order 3 and  $AB = 0AB = 0$ , then  $A = 0A = 0$ .

1 point

If  $AA$  is a square matrix of order 22 whose first column is denoted by  $C_1$  and second column is denoted by  $C_2$  (the left most column is taken as the first one) and  $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$  then choose the set of correct options.

[Hint: Recall the definition of matrix multiplication in terms of rows and columns.]



The first column of  $ABAB$  is  $b_{11}C_1 + b_{21}C_2$ .



The second column of  $ABAB$  is  $b_{12}C_1 + b_{22}C_2$ .



The first column of  $ABAB$  is  $b_{21}C_1 + b_{11}C_2$ .



The second column of  $ABAB$  is  $b_{22}C_1 + b_{21}C_2$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

The first column of  $ABAB$  is  $b_{11}C_1 + b_{21}C_2$ .

The second column of  $ABAB$  is  $b_{12}C_1 + b_{22}C_2$ .

1 point

Choose the set of correct options.

[Hint: Assume  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$  and use the given condition to find matrix  $AA$  for last two options.]



There exist some real matrices  $AA$  and  $BB$ , such that  $AB = BAAB = BA$ .



There do not exist any real matrices  $AA$  and  $BB$ , such that  $AB = BAAB = BA$ .



There does not exist any real matrix  $AA$ , such that  $A^2 = AA^2 = A$ .



There exists some real  $2 \times 2 \times 2$  matrix  $AA$ , such that  $A^2 + A + I = 0A^2 + A + I = 0$

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

There exist some real matrices  $AA$  and  $BB$ , such that  $AB = BAAB = BA$ .

There exists some real  $2 \times 2 \times 2$  matrix  $AA$ , such that  $A^2 + A + I = 0A^2 + A + I = 0$

Suppose  $AA$  is a  $3 \times 3$  scalar matrix and (1,1)-th entry of the matrix  $AA$  is 4. Suppose  $BB$  is a  $3 \times 3$  square matrix such that  $(i, j)$ -th entry is equal to  $i^2 + j^2$ . Find the (2,2)-th entry of the

matrix  $2A + B$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**

(Type: Numeric) 16

1 point

Suppose  $A = \begin{bmatrix} -1 & 2 \\ 4 & 7 \end{bmatrix}$  and  $A^2 - \alpha A + I = 0$  for some  $\alpha \in \mathbb{R}$ .

Find the value of  $\alpha$ .

**Yes, the answer is correct.**

**Score: 1**

**Accepted Answers:**